Deviation report — open at initiation

Deviation number	DEV-2026-1184
Title	RS-3 and assay out of specification after crystallization excursion
Date/time of occurrence	12 March 2026, 14:40 IST
Date/time of detection	14 March 2026, 09:15 IST, during QC HPLC review
Site	Northline API Works, Hyderabad
Area	Crystallization bay 2
Equipment	Crystallizer CR-300; cooling-water loop CW-02; valve V-17
Product	Atorvastatin calcium, crystalline API
Batch / lot	ATV-CA-26-0142
Process stage	Final crystallization and isolation
Proposed classification	Major — specification failure, batch quarantined, not shipped
Raised by	L. Joseph, production supervisor, B shift
Received by QA	14 March 2026, 09:40 IST

Description

QC reported related substance RS-3 (desfluoro impurity) at 0.18% against a specification of NMT 0.15% for atorvastatin calcium batch ATV-CA-26-0142. Assay on the anhydrous basis was 98.2% against 98.5–101.5%. Both results are out of specification. The batch was set to Quarantine in the inventory system and has not been shipped. It is intended for EU CEP supply. OOS procedure QA-SOP-014 has been initiated. The previous batch from the same crystallizer, ATV-CA-26-0141, met RS-3 at 0.09%.

The batch record shows that during the crystallization hold the jacket temperature rose from the target of 8°C to 17°C and stayed there for 46 minutes, from 14:22 to 15:08 on 12 March 2026. Operator R. Menon wrote in the B-shift log: cooling water pressure fluctuating, V-17 hunting. Maintenance work order WO-88421 was open on the CW-02 strainer. The strainer was cleaned on 11 March, but the differential-pressure alarm was left suppressed in the DCS until 13 March 2026 at 06:00.

Results at detection

Test	Result / specification / call
Related substance RS-3	0.18% / NMT 0.15% / OOS
Assay, anhydrous basis	98.2% / 98.5–101.5% / OOS
Previous batch ATV-CA-26-0141 RS-3	0.09% / NMT 0.15% / conforms

Immediate actions

- Batch ATV-CA-26-0142 status set to Quarantine. No material has been shipped.
- Crystallizer CR-300 tagged out of service pending an engineering check of valve V-17 and loop CW-02.
- Retained samples pulled. Mother-liquor samples from the crystallization have been requested and are not yet tested.
- HPLC worksheet held with the batch record. Analyst noted manual integration of the RS-3 peak.

Facts recorded at initiation

- Solvent lot IPA-8891 is the same lot used for batch ATV-CA-26-0141 and for two earlier conforming batches. The CoA conforms.
- Seed lot SL-77 CoA conforms. The charge was recorded as 1.2 kg against a batch-record target of 1.5 kg. The weighing printout from balance BAL-QC-04 is missing from the batch record. A second-person verification signature is present but illegible.
- HPLC method STM-ATV-RS revision 7. System suitability passed. The analyst noted that the RS-3 standard was added mid-sequence because it had not arrived when the SST sequence was started. Integration of RS-3 was manual.

- During isolation, cleanroom humidity was 62% against a limit of NMT 55% for about 3 hours, coinciding with an AHU-4 filter change. Isolation was after the crystallization excursion.

Impact noted by production — QA to confirm

Product quality: two specification failures on the release tests. The batch cannot be released on these results. Patient safety: no patient exposure is identified; the batch has not left the site and has not been administered. Regulatory: the batch was planned for EU CEP supply, so a confirmed OOS must be investigated under QA-SOP-014 before any disposition. Batch disposition at the time of this report: Quarantine.

Not yet established

- Whether the RS-3 peak is confirmed on reinjection with a complete system-suitability sequence, including the RS-3 standard present from the start.
- Who suppressed the CW-02 differential-pressure alarm, and the DCS audit trail for that change.
- The maintenance history of valve V-17, beyond the note that it was hunting.
- The missing balance printout for the seed charge on BAL-QC-04.
- A formal interview with operator R. Menon.

This report records the deviation at initiation. It does not conclude a root cause. Investigation method to be agreed by QA: 5 Whys and an Ishikawa diagram covering Man, Machine, Method, Material, Measurement, and Environment.